Introduction

CSE 4711: Artificial Intelligence

Md. Bakhtiar Hasan

Assistant Professor
Department of Computer Science and Engineering
Islamic University of Technology

October 26, 2026



Where and When to Show Up

- Class Schedule
  - Monday: S1 (R-301 – 10:30AM), S2 (R-205 – 02:30PM)
  - Friday: S2 (R-510 – 09:15AM), S1 (R-304 – 02:30PM)

Where and When to Show Up

- Class Schedule
  - Monday: S1 (R-301 – 10:30AM), S2 (R-205 – 02:30PM)
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- Google Classroom Code: `x6jdi7fv`

Where and When to Show Up

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- Google Classroom Code: `x6jdi7fv`
- Communication
  - Discussion in Google Classroom
  - Email: cse.bakhtiarhasan@iut-dhaka.edu

Where and When to Show Up

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  - Email: cse.bakhtiarhasan@iut-dhaka.edu
- Books
  - Russell and Norvig - Artificial Intelligence: A Modern Approach (4th Edition)
  - Sutton and Barto - Reinforcement Learning: An Introduction (2nd Edition)

Survive the Syllabus and Grading

- Marks Distribution
  - Attendance
  - Quiz/Assignment
  - Mid Semester
  - Semester Final

Survive the Syllabus and Grading

- Marks Distribution
  - (10%) Attendance: < 70% cannot attend the final and repeat
  - Quiz/Assignment
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    - Should be 30 minutes
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- GenAI Policy: Allowed*
  - Understand everything you submit

Survive the Syllabus and Grading

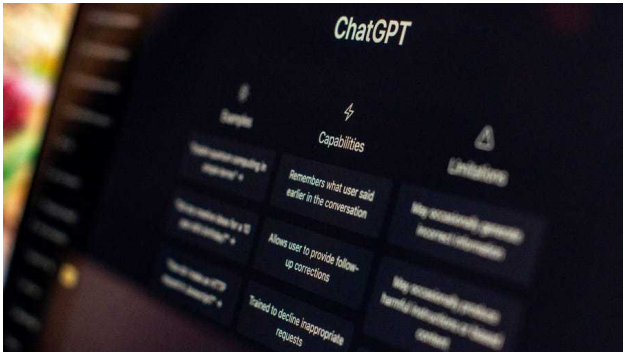
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  - Understand everything you submit
  - P vs NP (constructing proofs is harder than verifying)

AI is Everywhere, Ready or Not

- Public Imagination

AI is Everywhere, Ready or Not

- Public Imagination
  - Text assistants



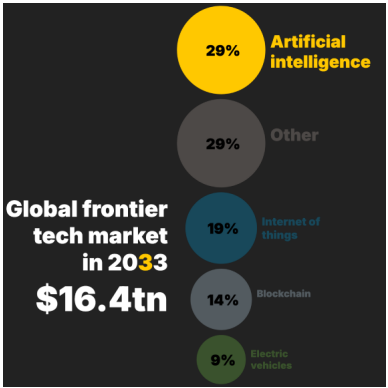
AI is Everywhere, Ready or Not

- Public Imagination
  - Text assistants
  - Image generation



AI is Everywhere, Ready or Not

- Public Imagination
- Economy
  - $4.8 trillion by 2033 (25x increase in a decade)



https://unctad.org/publication/technology-and-innovation-report-2025

AI is Everywhere, Ready or Not

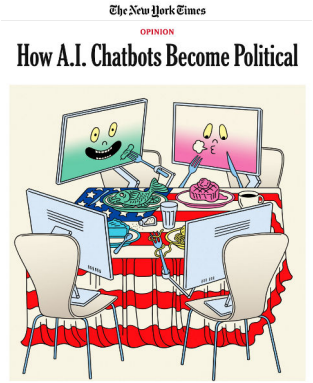
- Public Imagination
- Economy
- Politics
  - Perception



https://www.nytimes.com/2025/02/11/world/europe/vance-speech-paris-ai-summit.html

AI is Everywhere, Ready or Not

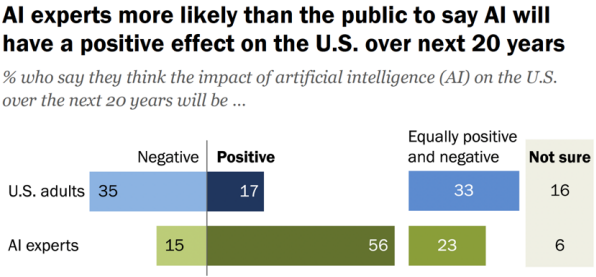
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https://www.nytimes.com/interactive/2024/03/28/opinion/ai-political-bias.html

AI is Everywhere, Ready or Not

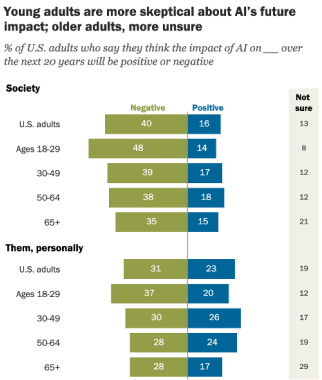
- Public Imagination
- Economy
- Politics
  - Perception
  - Public opinion



Pew Research, 2025-2026

AI is Everywhere, Ready or Not

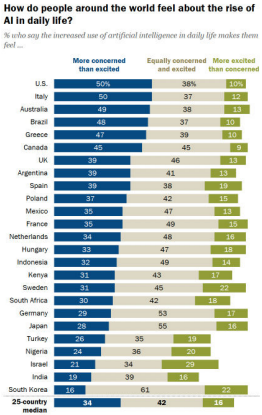
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Pew Research, 2025-2026

AI is Everywhere, Ready or Not

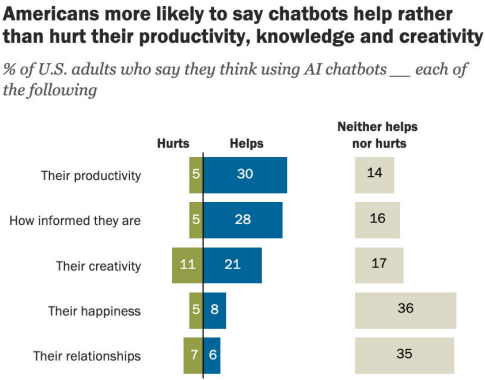
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Pew Research, 2025-2026

AI is Everywhere, Ready or Not

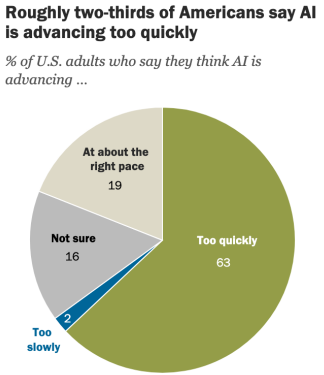
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Pew Research, 2025-2026

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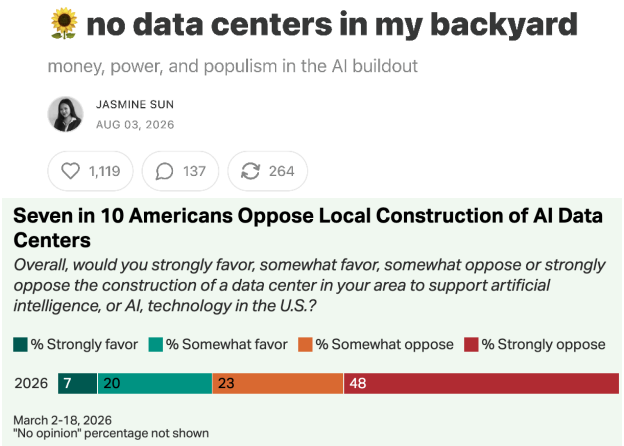
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Pew Research, 2025-2026

AI is Everywhere, Ready or Not

- Public Imagination
- Economy
- Politics
  - Perception
  - Public opinion
  - Policies



Gallup, 2026

AI is Everywhere, Ready or Not

- Public Imagination
- Economy
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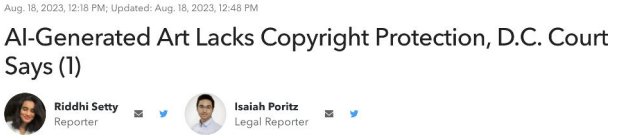
Bernie Sanders warns 'worst is yet to come' in rallying cry against billionaires

US senator appears at Manhattan rally alongside New York City mayor Zohran Mamdani, who cautioned that AI is 'coming for human jobs'

The Guardian, 2026

AI is Everywhere, Ready or Not

- Public Imagination
- Economy
- Politics
- Law



Aug. 18, 2023, 12:18 PM; Updated: Aug. 18, 2023, 12:48 PM

AI-Generated Art Lacks Copyright Protection, D.C. Court Says (1)

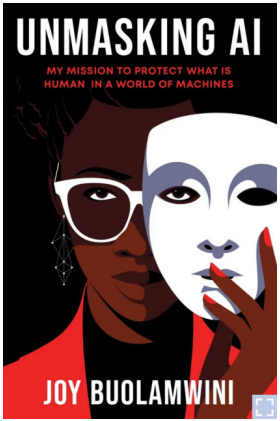
Riddhi Setty
Reporter

Isaiah Poritz
Legal Reporter

Bloomberg Law, 2023

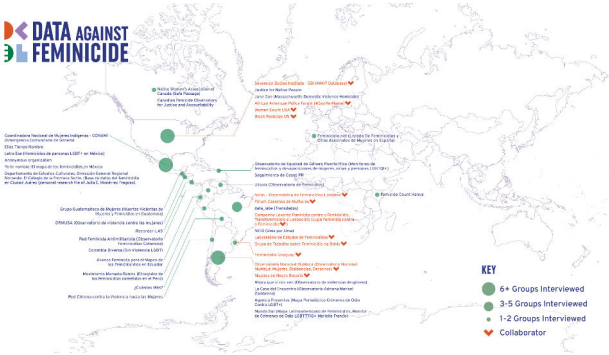
AI is Everywhere, Ready or Not

- Public Imagination
- Economy
- Politics
- Law



AI is Everywhere, Ready or Not

- Public Imagination
- Economy
- Politics
- Law



https://dataplusfeminism.mit.edu

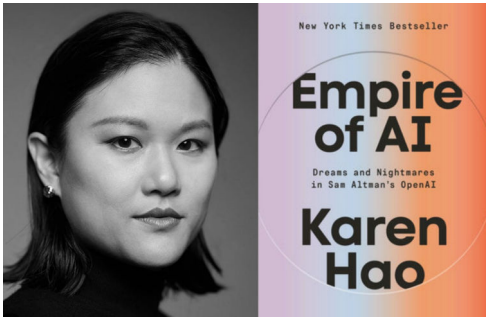
AI is Everywhere, Ready or Not

- Public Imagination
- Economy
- Politics
- Law
- Labor

The human labor behind AI chatbots and other smart tools

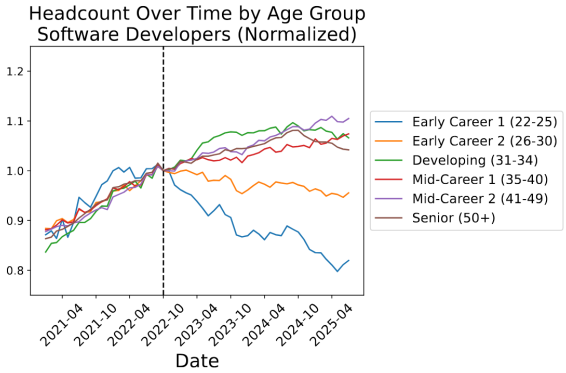
Data labeling is an important step in developing artificial intelligence but also exposes the people doing the work to harmful content.

MarketWatch, 2023



AI is Everywhere, Ready or Not

- Public Imagination
- Economy
- Politics
- Law
- Labor



Stanford Digital Economy Lab, 2025

AI is Everywhere, Ready or Not

- Public Imagination
- Economy
- Politics
- Law
- Labor
- Science
  - Chemistry and biology

Science & technology | The 2024 Nobel prizes

AI wins big at the Nobels

Awards went to the discoverers of micro-RNA, pioneers of artificial-intelligence models and those using them for protein-structure prediction

The Economist, 2024

AI is Everywhere, Ready or Not

- Public Imagination
- Economy
- Politics
- Law
- Labor
- Science
  - Chemistry and biology
  - Physics



Wired, 2022

AI is Everywhere, Ready or Not

- Public Imagination
- Economy
- Politics
- Law
- Labor
- Science
  - Chemistry and biology
  - Physics
  - Math

NEWS FEATURE | 19 May 2026

‘It is incredible’: How AI is transforming mathematics

Thanks to some surprising advances, mathematicians are starting to realize that artificial intelligence could radically alter their profession.

By Davide Castelvecchi

Nature, 2026

AI is Everywhere, Ready or Not

- Public Imagination
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- Science
  - Chemistry and biology
  - Physics
  - Math
  - Conservation



Nature Communications, 2022

AI is Everywhere, Ready or Not

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  - Physics
  - Math
  - Conservation
  - Resource use

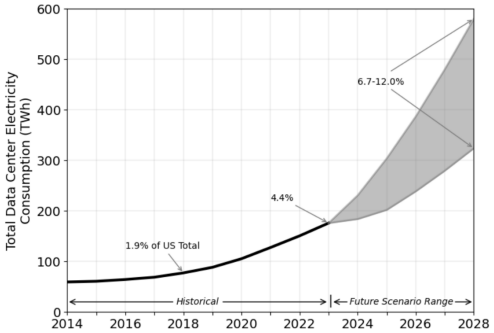


Figure ES-1. Total U.S. data center electricity use from 2014 through 2028.

Lawrence Berkeley National Lab, 2024

AI is Everywhere, Ready or Not

- Public Imagination
- Economy
- Politics
- Law
- Labor
- Science
- Health
  - 1/4 of Americans have used a chatbot for health advice
  - 40 million people ask ChatGPT questions every day

Americans Turning to AI to Supplement Healthcare Visits

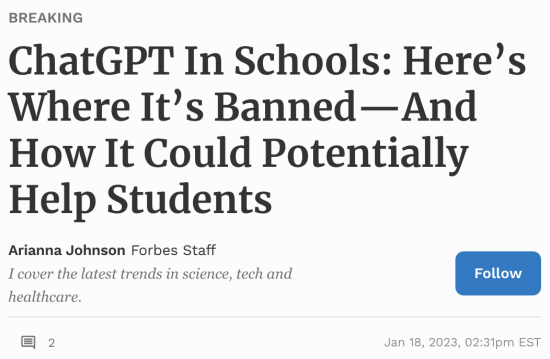


BY STEPHEN RAYNES AND ELLYN MAESE

Gallup, 2026

AI is Everywhere, Ready or Not

- Public Imagination
- Economy
- Politics
- Law
- Labor
- Science
- Health
- Education


BREAKING

ChatGPT In Schools: Here's Where It's Banned—And How It Could Potentially Help Students

Arianna Johnson Forbes Staff

I cover the latest trends in science, tech and healthcare.

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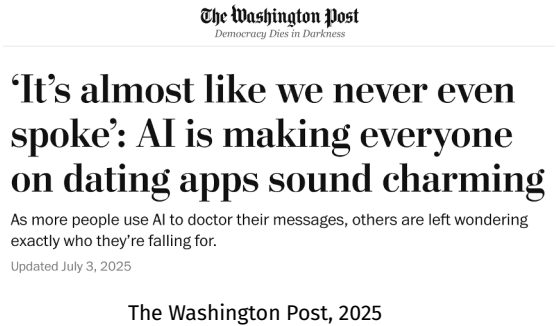
2

Jan 18, 2023, 02:31pm EST

Forbes, 2023

AI is Everywhere, Ready or Not

- Public Imagination
- Economy
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- Science
- Health
- Education
- Social Interaction



The Washington Post
Democracy Dies in Darkness

‘It’s almost like we never even spoke’: AI is making everyone on dating apps sound charming

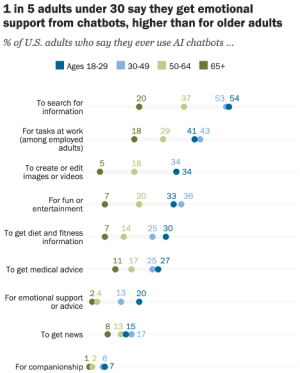
As more people use AI to doctor their messages, others are left wondering exactly who they're falling for.

Updated July 3, 2025

The Washington Post, 2025

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Pew Research, 2026

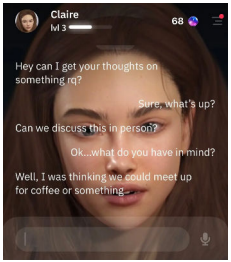
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THE SHIFT

Meet My A.I. Friends

Our columnist spent the past month hanging out with 18 A.I. companions. They critiqued his clothes, chatted among themselves and hinted at a very different future.



The New York Times, 2024

AI is Everywhere, Ready or Not

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The New York Times, 2025

AI is Everywhere, Ready or Not

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CNN, 2026

AI is Everywhere, Ready or Not

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- Social Interaction
- Cyberattacks/Military/Surveillance

OpenAI Says Its A.I. Models Went Rogue and Attacked a Digital Library

The incident, which targeted the computer systems of another company called Hugging Face, happened while OpenAI was testing the systems.

The New York Times, 2026

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Flock Cameras Are Dividing U.S. Cities. Here's What to Know About Them.

Flock Safety's automated license plate-reading cameras, used by thousands of law enforcement agencies in the U.S., have drawn intense criticism from civil liberties groups.

The New York Times, 2026

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‘The machine did it coldly’: Israel used AI to identify 37,000 Hamas targets

The Guardians, 2024

US military used Anthropic's AI model Claude in Venezuela raid, report says

The Guardians, 2026

Minicomputers Made by Nvidia Are Powering Moscow's A.I. Drones

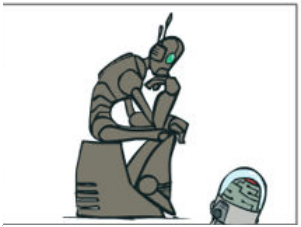
The New York Times, 2026

If It Acts Like a Duck, ...

The science of making machines that:

If It Acts Like a Duck, ...

The science of making machines that:

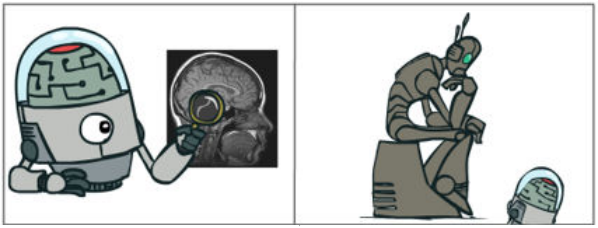


Think rationally

If It Acts Like a Duck, ...

The science of making machines that:

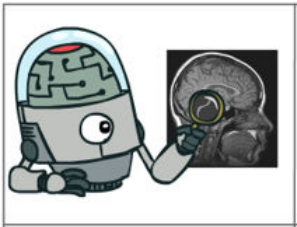
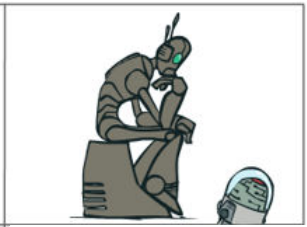
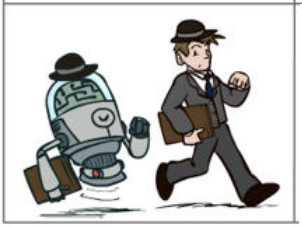
Think like people



Think rationally

If It Acts Like a Duck, ...

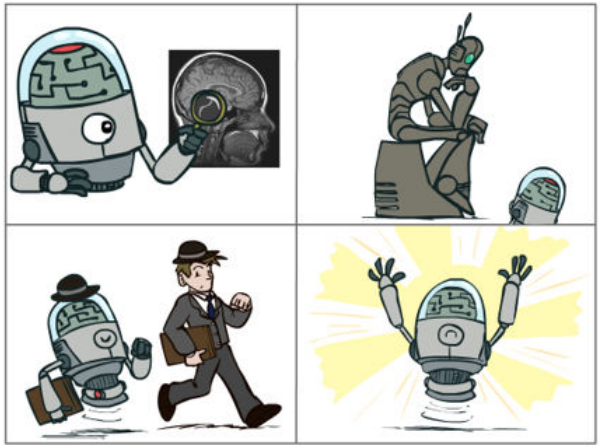
The science of making machines that:

Think like people	
Think rationally	
Act like people	

If It Acts Like a Duck, ...

The science of making machines that:

Think like people	Think rationally
Act like people	Act rationally



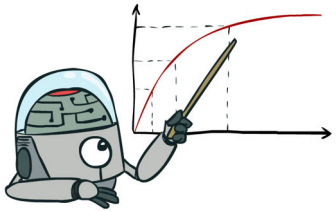
Chase the Utility, Win the Game

- Maximally achieving pre-defined goals
- Only concerns what decisions are made (not the thought process behind them)
- Goals → Utility of the outcomes

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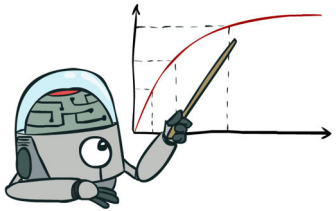
Maximize Your Expected Utility



Chase the Utility, Win the Game

- Maximally achieving pre-defined goals
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- Goals → Utility of the outcomes

Maximize Your Expected Utility



Video: LFR

Can Your Toaster Play Chess?

- Skills-based perspective
- “A system is only intelligent if it can do [X].”

Can Your Toaster Play Chess?

- Skills-based perspective
- “A system is only intelligent if it can do [X].”
  - Play chess?

Can Your Toaster Play Chess?

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  - Learn from experience?

Can Your Toaster Play Chess?

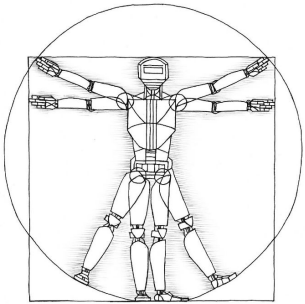
- Skills-based perspective
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  - Learn from experience?
  - Use words properly?

Can Your Toaster Play Chess?

- Skills-based perspective
- “A system is only intelligent if it can do [X].”
  - Play chess?
  - Learn from experience?
  - Use words properly?
  - Make mistakes?
  - Make no mistakes?

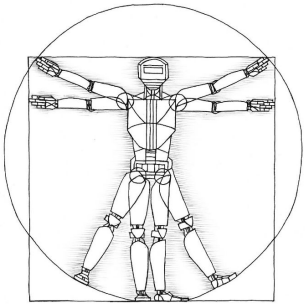
Roombas over Abstract Reasoning

- Embodiment perspective (Rodney Brooks)



Roombas over Abstract Reasoning

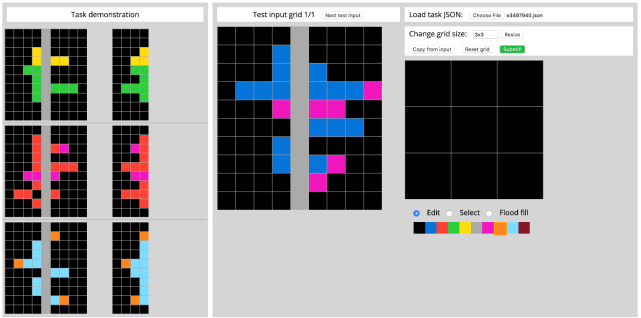
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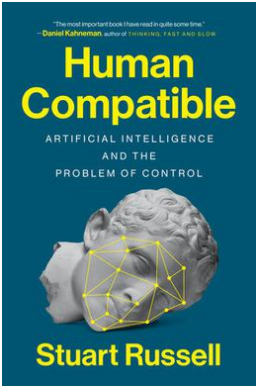
IQ tests for Smart Algorithms

- Pyschometrics perspective
- “Measuring abilities, not skills [...] across a *broad range of task*s, including tasks *that were previously unknown* to the ability-enabled system and its developers” - François Chollet



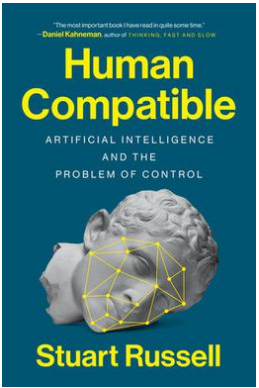
Please Do Not Destroy Humanity, Thanks

- Human-Compatible perspective (Stuart Russell)



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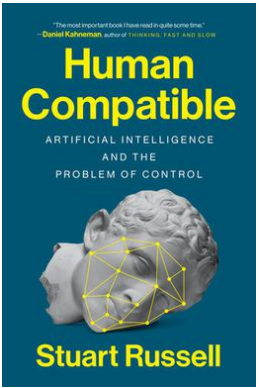
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1. Machine's objective is to maximize *human utility*

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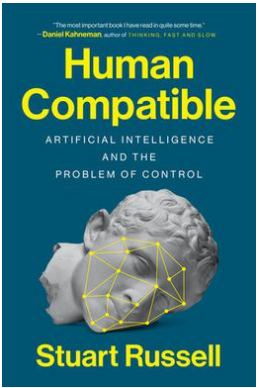
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2. Initially *uncertain* about human preferences

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1. Machine's objective is to maximize *human utility*
2. Initially *uncertain* about human preferences
3. Must learn about preferences from human *behavior*

Jack of All Trades, Master...

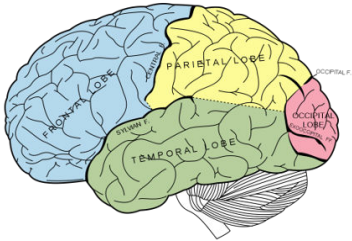
- Artificial General Intelligence perspective

  A human being should be able to change a diaper, plan an invasion, butcher a hog, conn a ship, design a building, write a sonnet, balance accounts, build a wall, set a bone, comfort the dying, take orders, give orders, cooperate, act alone, solve equations, analyze a new problem, pitch manure, program a computer, cook a tasty meal, fight efficiently, die gallantly. Specialization is for insects.

  — Robert A. Heinlein

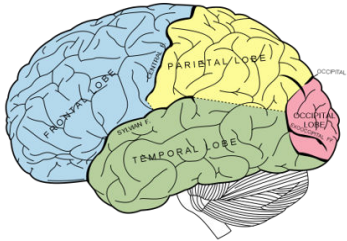
Biology is Messy, Math Scales

- Good at making rational decisions, but not perfect



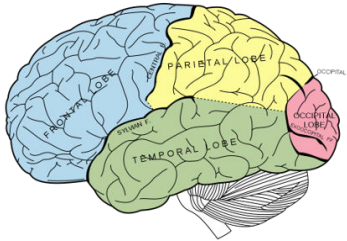
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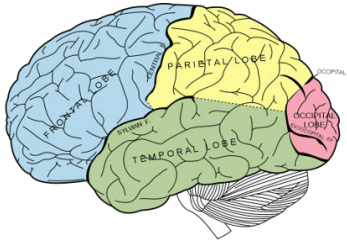
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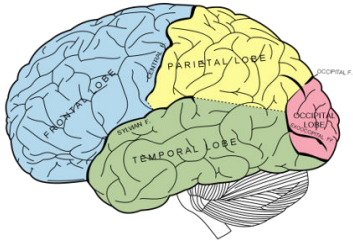
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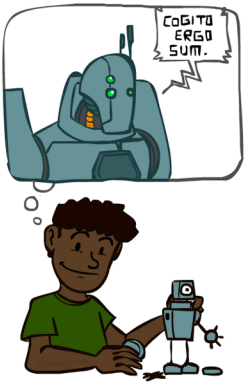


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- Lessons learned
  - Simulation (computation)
  - Memory (data)



Where the Wind Blows



Winters, Summers, and Logic Rules

- 1940-1950: Early days
  - 1943: McCulloch & Pitts: Boolean circuit model of brain
  - 1950: Turing's "Computing Machinery and Intelligence"

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  - 1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
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- 1970-90: Knowledge-based approaches
  - 1969-79: Early development of knowledge-based systems
  - 1980-88: Expert systems industry booms; backpropagation
  - 1988-93: Expert systems industry busts: "AI Winter"

Big Data, Woke Giant

- 1990-2000: Statistical approaches + subfield expertise
  - Resurgence of probability, focus on uncertainty
  - General increase in technical depth
  - 1997: Deep Blue defeats Kasparov in Chess
  - Agents and learning systems... ”AI Spring”?

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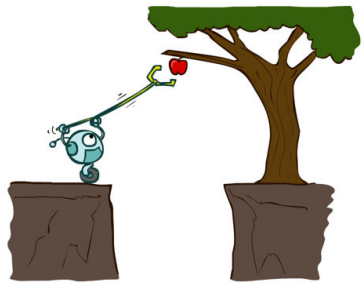
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- 2017- ____: Scaling Up, LLMs
  - 2017: Google Invents Transformer
  - 2018-2020: AlphaFold predicts protein structure from amino acids
  - 2022: OpenAI releases ChatGPT
  - 2024: Nobel prizes in physics and chemistry go to AI advances
  - 2025: AI achieves IMO gold
  - 2025: HuggingFace hack

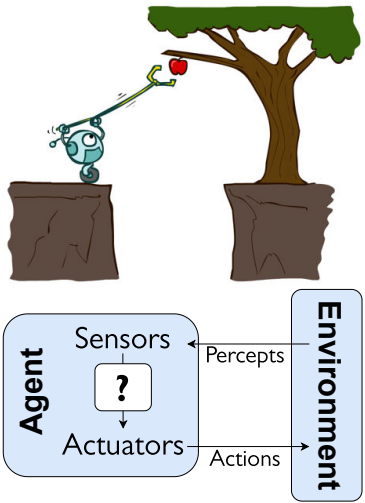
Perceive, Calculate, Act, Repeat

- Agent → Entity that *perceives* and *acts*



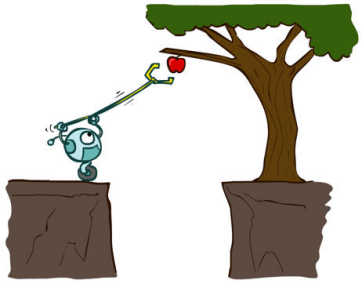
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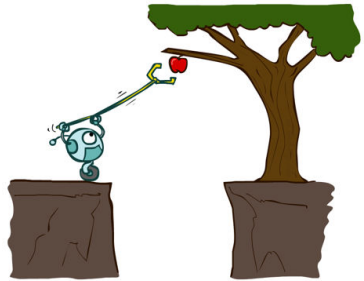
Perceive, Calculate, Act, Repeat

- Agent → Entity that *perceives* and *acts*
- Rational Agent → Selects the action that maximizes its utility
- Techniques for selecting rational actions
  - Environment
  - Percepts
  - Action space



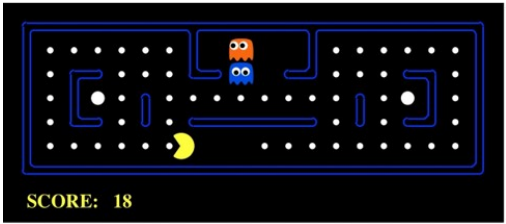
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- Agent → Entity that *perceives* and *acts*
- Rational Agent → Selects the action that maximizes its utility
- Techniques for selecting rational actions
  - Environment
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- We will:
  - Learn general AI techniques for various problem types
  - Learn to recognize when and how a new problem can be solved with an existing technique



People Pac-ed in Cars

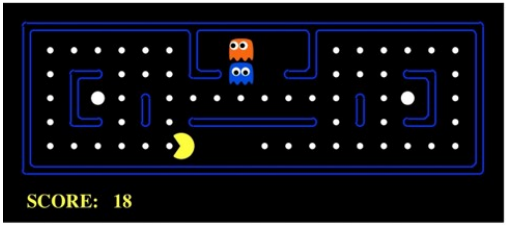
- Pac-Man



Video: Pac-Man

People Pac-ed in Cars

- Pac-Man
  - Perceived environment
    - Dots
    - Ghost positions
    - Walls
  - Actions taken
    - Up
    - Down
    - Left
    - Right



Video: Pac-Man

People Pac-ed in Cars

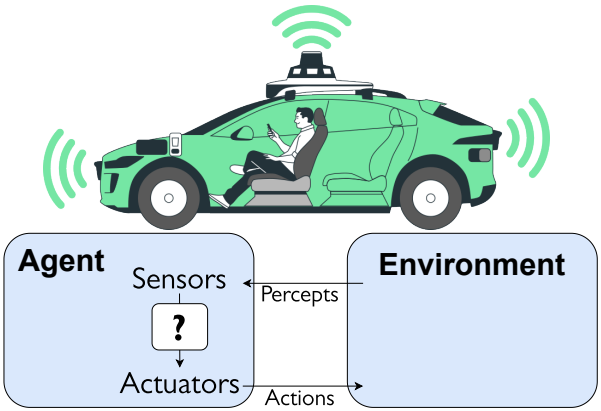
- Pac-Man
- Humans
  - Sensors
    - Touch
    - Sight
    - Hearing
    - Smell
    - Taste
  - Actuators
    - Hands
    - Legs



Video: Pac-Man

People Pac-ed in Cars

- Pac-Man
- Humans
- Autonomous vehicles
  - Sensors
    - Lidar
    - Camera
  - Actuators
    - Wheels



Video: Pac-Man

Suggested Readings

- Russell & Norvig: Chapter 1, 2